

A unital singleton model acts in finite dimension

MathlibAnnex.Analysis.CStarAlgebra.Representation.finiteDimensional_space_of_singleton

theorem

Uses simplicity to put the identity operator in the compact image ideal.

Statement

Let A be a nonzero unital complex C^* -algebra and let $\pi : A \rightarrow \mathcal{B}(H)$ be a nonzero irreducible unital $*$ -representation. Assume every nonzero irreducible unital $*$ -representation of A is unitarily equivalent to π . Then H is finite dimensional over $\mathbb{C}$.

Assumptions

The uniqueness-of-class condition is essential to the argument; mere irreducibility is not asserted to force finite dimension. Separability is assumed for H , not for A .

Conclusion

The representation space has a finite complex basis.

Proof route

Obtain faithfulness and simplicity, find a nonzero compact image, and use the resulting nonzero closed ideal to make the identity compact.

▼ Proof steps

1. Apply Faithfulness of a unital singleton representation to the given nonzero ordered unital A and singleton π to obtain injectivity. Then Simplicity from a faithful singleton model applies with that injectivity and shows that every closed two-sided ideal of A is 0 or A .
2. Use A scalar-corner projection in the unital algebra with this singleton representation on separable H to obtain a nonzero projection $p \in A$ with $pap \in \mathbb{C}p$ for every $a \in A$. Injectivity gives $\pi(p) \neq 0$. Irreducibility, this nonzero image, and the scalar-corner equations are precisely the inputs of A nonzero represented scalar corner is compact, whose output is compactness of $\pi(p)$.
3. Let $I = \{a \in A : \pi(a) \text{ is compact}\}$, the two-sided ideal of The compact-preimage ideal. It is norm closed by Closedness of the compact-preimage ideal. Since $p \in I$ and $p \neq 0$, $I \neq 0$; simplicity therefore forces $I = A$. In particular $1_A \in I$.
4. By the definition of I , $\pi(1_A)$ is compact. Unitality now matters:

$$\pi(1_A) = I_H \in \mathcal{K}(H).$$

The Compact-identity criterion applied to the same H gives $\dim_{\mathbb{C}} H < \infty$, as used in the final source step.

Main citations

- Exact theorem and proof
- Faithfulness of a unital singleton representation
- Closed-ideal simplicity
- A scalar-corner projection in the unital algebra
- A nonzero represented scalar corner is compact
- Compact-preimage ideal
- Closedness of the compact-preimage ideal

Lean source signature (exact)

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theorem finiteDimensional_space_of_singleton [Nontrivial A]
  [TopologicalSpace.SeparableSpace H]
  (pi : Representation A H)
  (hsingle : IsSingletonIrreducibleModel.{u, v, u} pi) :
  FiniteDimensional  $\mathbb{C}$  H
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In the source	Mathematical meaning
[Nontrivial A]	The algebra is nonzero: $A \neq \{0\}$. This condition does not say whether a unit is assumed; that information comes from the surrounding C^* -algebra structure.
[TopologicalSpace.SeparableSpace H]	The complex Hilbert space H is separable.
(pi : Representation A H)	The unital representation π , so $\pi(1_A) = I_H$.
(hsingle : IsSingletonIrreducibleModel.{u, v, u} pi)	π is nonzero irreducible and is unitarily equivalent to every nonzero irreducible unital comparison representation.
FiniteDimensional $\mathbb{C}$ H	The whole space H has finite dimension over the complex numbers, not just some invariant subspace.

Exact source and proof.

Earlier published Card and PDF

▼ Exact content identity

Declaration: MathlibAnnex.Analysis.CStarAlgebra.Representation.finiteDimensional_space_of_singleton

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